香港科技大学(广州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

UCUG1903 Signal and Information Technology

L01 Part 1 Course Introduction

Dr. Shijian GAO

Acknowledgement

Andrew Wing On POON
潘 永 安
Department Head and Professor

Kai Sun Albert WONG
王 啓 新
Associate Head for UG Studies and Outreach and Senior Lecturer

Roger Shu-Kwan CHENG
鄭 樹 坤
Professor Emeritus

GEORGE JIE YUAN 袁 杰

PhD in Electrical Engineering

Professor
Associate Dean of Engineering (Research & Graduate Studies)

AU YEUNG Siu Kei
Senior Lecturer, Electronic Engineering and Computer Science

Email
jauyeung@hkmu.edu.hk

Xuning ZHANG
张栩宁

PhD in Electronic and Computer Engineering
The Hong Kong University of Science and Technology, 2017

Lecturer I

Pillar of STEM Education

Assistant Professor of Practice in Engineering Education

Pillar of STEM Education

Course Information

- **Instructor**: Dr. Shijian Gao
- **Email**: shijiangao@hkust-gz.edu.cn
- **Office Hour**: Friday 2:00pm-2:45pm, W1-410
- **Time & Location**: Friday 15:00-17:50, E1-134

- TA: Mr. Gangyong Zhu and Mr. Jiahui Liang
- Email: gzhu097@connect.hkust-gz.edu.cn & jliang097@connect.hkust-gz.edu.cn

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Warm-Up Self-Introduction

- Your Grade
- Your Major (Intended)
- Why do you register this course?
- Do you have any linear algebra background?
- Do you have a basic mastery of complex numbers and their calculations?
- Programming language: Python, Matlab

Power of Information Technology in A Single Handheld Device

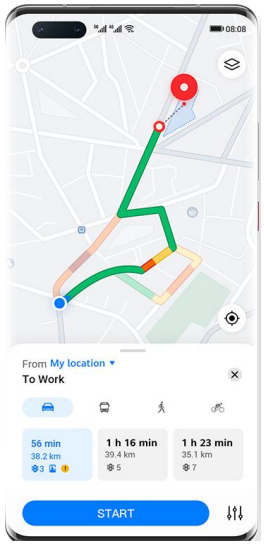
Smart Scan
(***Objects/Finger print/Iris/Face Recognition***)

5G Mobile
(***Mobile communications***)

e.g.
Huawei
Phone

Satellite Calls
(***Satellite Mobile Communication System, Microchip Technology, Signal Amplification***)

Huawei Pay
(***Near Field Communications***)



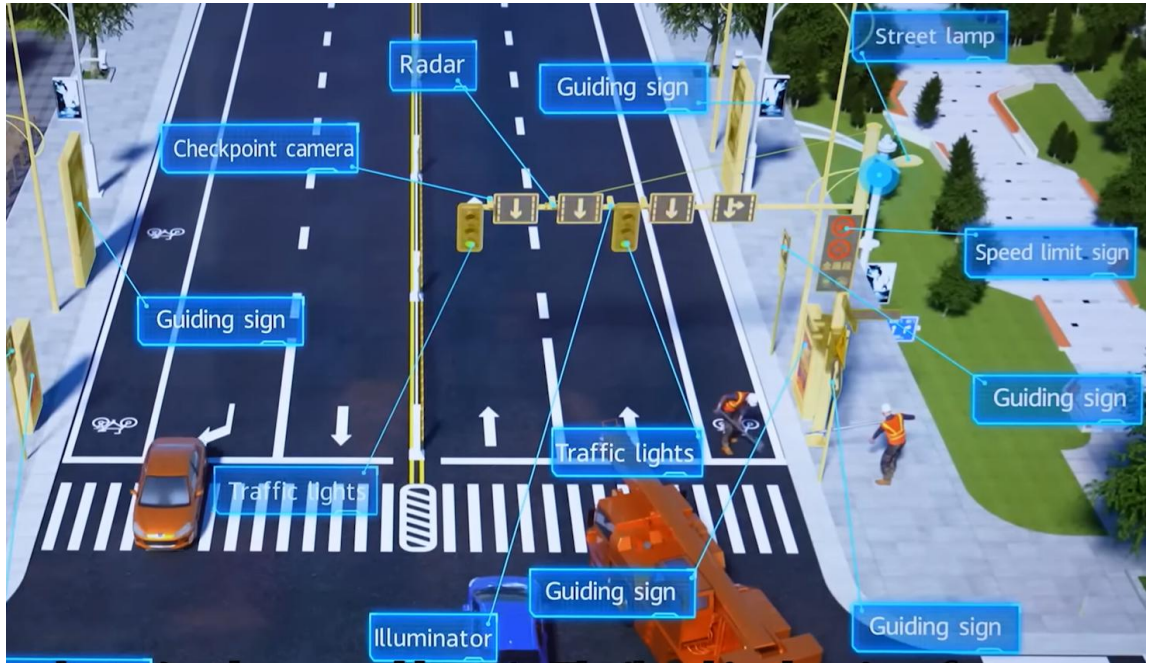
Supper Global Navigation Satellite System
(***Satellite Positioning and Navigation Technology***)

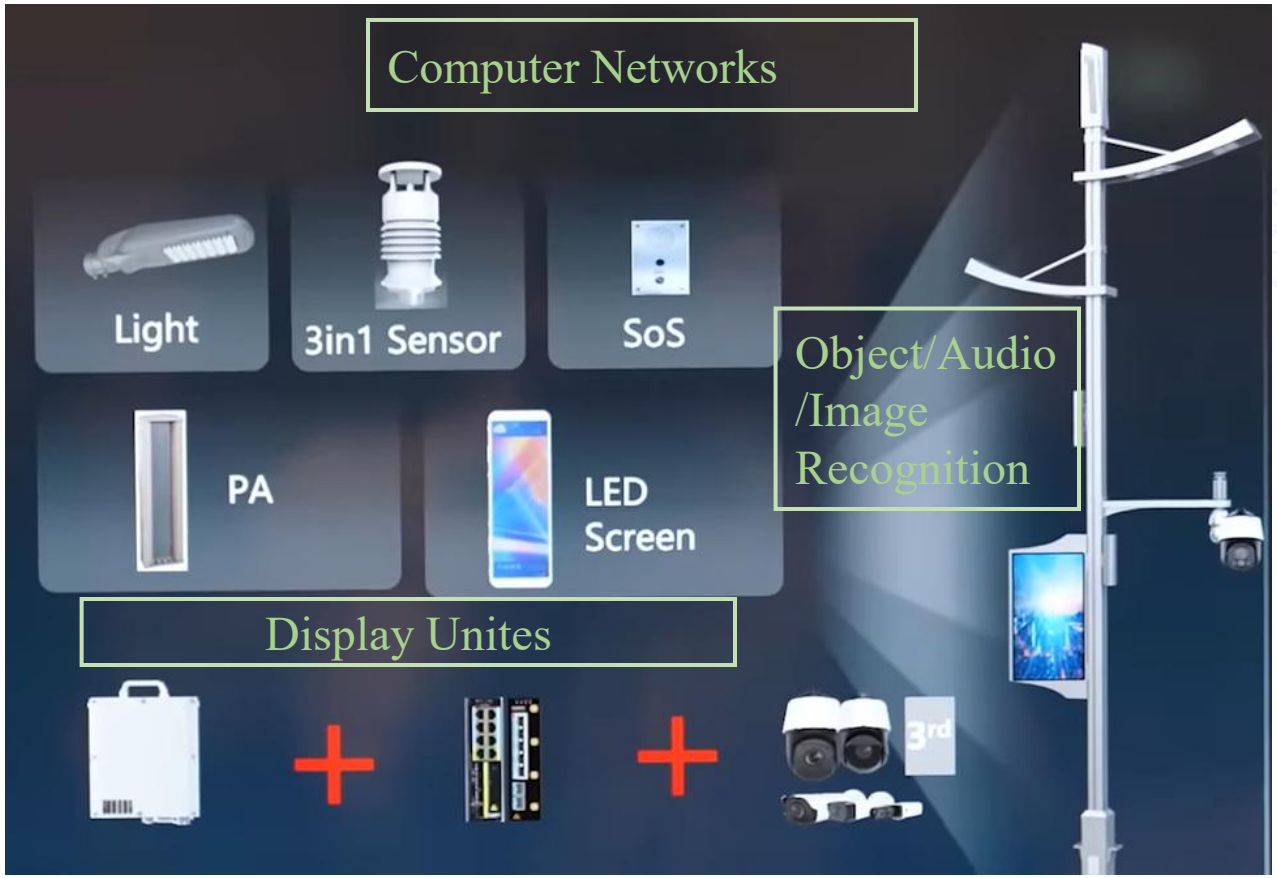
Voice Assistant
(***Digital Audio Recognition/Matching***)

Camera
(***Digital Image and Video Processing***)

Pictures are from https://www.gsmarena.com/huawei_mate_60_pro-12530.php, https://consumer.huawei.com/za/community/details/Huawei-Mate-60-Pro-is-the-world-s-first-satellite-calling-phone/topicId_280286/, https://www.octopus.com.hk/tc/consumer/mobile-payment/huawei-pay/about/index.html

Huawei City Management System

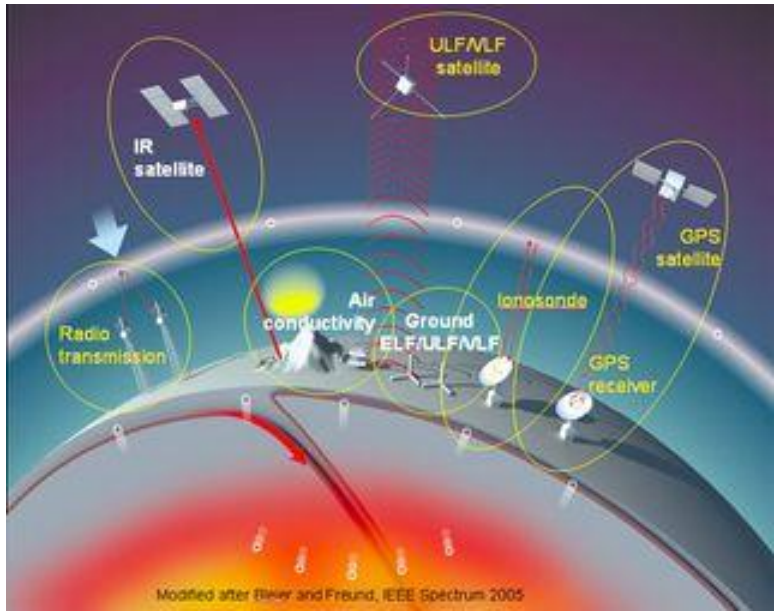




Pictures are from https://weibo.com/7476889887/M7Sb2ayOB

Signal and Information Technology & Us

- Signals and signal processing are pervasive in our world and enable many of our everyday experiences.
- Information technology has made the education process more effective and productive.
- Information technologies (IT) are drivers of productivity and the economies.
- Information technologies often enable scientific and technological progresses in many disciplines (e.g. broadband Internet enables tele-medicine; wireless sensor networks enables earthquake and tsunami detection).
- Huge investment and job market $$$.



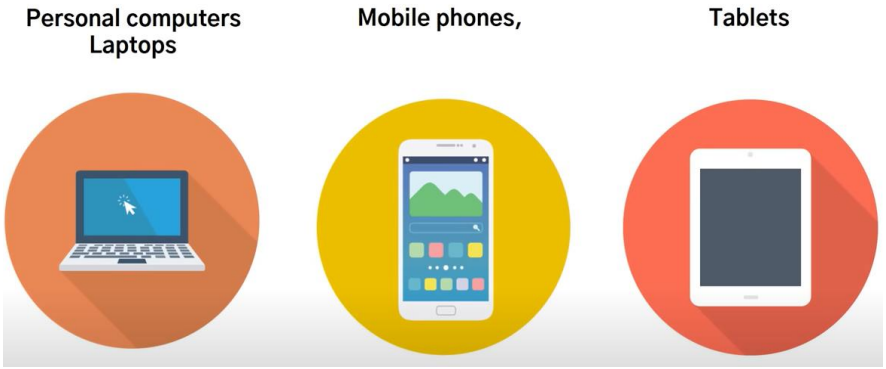
Pictures are from https://www.news-medical.net/health/What-is-Telemedicine.aspx, https://feweek.co.uk/toxic-online-learning-can-be-better-than-the-classroom/, https://www.space.com/6608-detecting-earthquakes-strike.html

What is Information Technology?

- **Information Technology Association of America (ITAA):** "the study, design, development, implementation, support or management of computer-based information systems, particularly software applications and computer hardware." IT deals with the use of electronic computers and computer software to convert, store, protect, process, transmit and retrieve information, securely.

- Sometimes the term “Information Technology” is also used to referred to everything that is high-tech.

Hardware



Software

Software systems

Applications

Pictures are from https://commons.wikimedia.org/wiki/File:HarmonyOS.jpg, https://www.youtube.com/watch?v=XZrckLYqdys

Course Materials

- No textbook
- There will be lecture notes, summary notes, some *optional* reading articles, and video tutorials from the Internet
- Homework, midterm and final exams will be based on the lecture notes and summary notes.

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Course Description

- This general education course introduces the basics of signals and information technology and their applications to daily-life consumer communication devices.
- ***BIG IDEAS*** include:
  - the representation of ***signals*** in the time and frequency domains;
  - ***digitization*** of information;
  - ***coding*** for data compression and error protection;
  - ***transmission*** of signals;
  - ***cellular*** networks and
  - ***multiple access*** technologies
- No prerequisite

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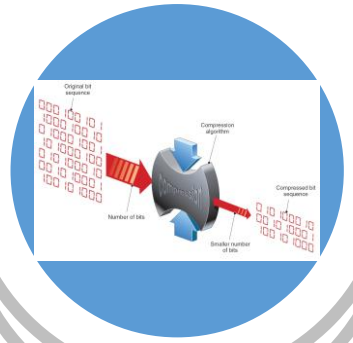
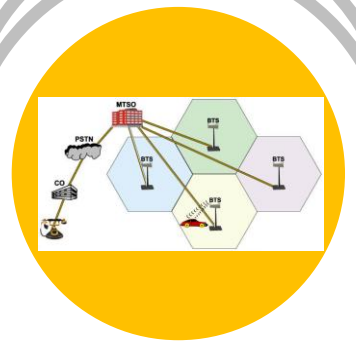
Course Outline

- The Course consists of 5 chapters

Digital Systems

Coding of Digital Information





Signals and Systems

Digitization

Communications

Course Outline

Chapter 2: Digital Systems
Represent signals in 0 and 1.

```
1110111011000110110101100011
1010011011110011001100010111
0011000100000101010111010011
1010010111111011111111100011
0001000111101101000110001100
0010111001111111011000101100
0100100010101101110110000100
```

Digitization

Chapter 3: Digitization
How to convert a continuous waveform to a sequence of 0 and 1.

Chapter 4: Coding of Digital Information
How to reduce information size?
How to protect information?

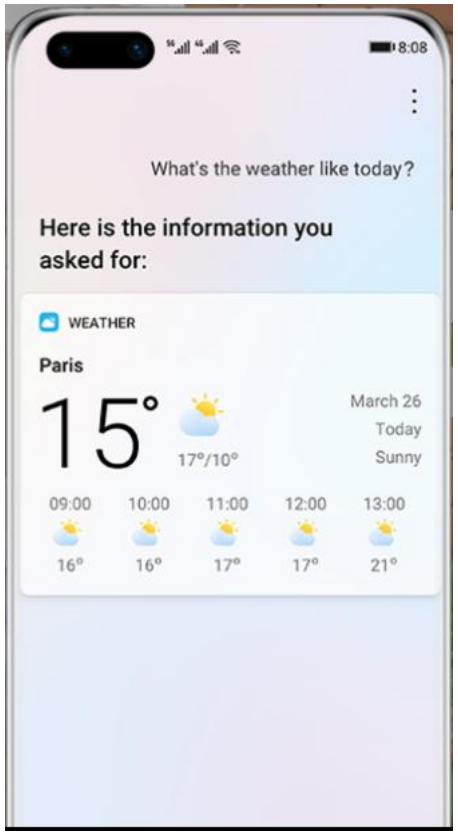
```
0010111001111111011000101100
0100100010101101110110000100
```

Matching with Phone's database

Bring your sunglasses, it's going to be nice in Paris.

Output signal

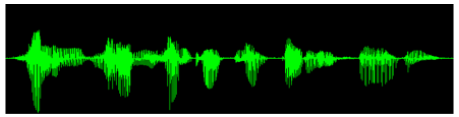
System



Huawei phone is a digital system

Input signal

What's the weather like today?



Chapter 1: Signals and Systems
How to represent a signal?
What components are contained in a signal?

Pictures are from https://consumer.huawei.com/en/emui/celia/, http://www.xinhuanet.com/world/2018-08/07/c_129927822.htm

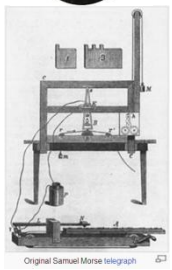
Course Outline

Chapter 5 Communications

How are phone signals transmitted?
AM and FM

Why multiple mobile users can make phone calls in the same cell at the same time?

Samuel Morse



Telegraph

Alexander Graham Bell

Telephone

Guglielmo Marconi

Long distance radio transmission

Hedy Lamarr

1G
Analog
system
FDMA

2G
Digital
system
FDMA
TDMA

3G
Spread
spectrum
technology

4G
OFDMA

5G
Optimized
OFDMA

Mobile phone then

Smart phone today

1844 | 1876 | 1901 | 1979 | 2009 | Now

Teaching and Learning Method – Blended Learning

Before lectures

The teaching team

Release course materials:

- Lecture notes
- Online articles
- Online videos

Students

- Learn online materials

During Lecture

- Further explanations to concepts and theorems
- Quiz
- Examples
- Demonstration
- In-class activities
  - Discussions
  - Worksheet

Exam

- Midterm
- Final

Project

- Presentation
- Report

Why blended learning?

- Provides personalized instruction
- Reinforces engagement, learning and retention
- Improved student learning outcomes

Grading

- Homework assignment: 20%
  - Four times, 5% for each
- Midterm Exam: 25% (Multiple choice, fill in blanks, computation and analytical problems)
- Final Exam: 35% (Multiple choice, fill in blanks, computation and analytical problems)
- Team project
  - Presentation: 10% (ppt slides + oral presentation)
  - Report: 10%

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Team Project

- Guidelines
  - A group of 3 members. All members of one group will get the same project score. Each group to propose one information technology topic based on their interest.
  - Project presentation slides (around 10-15 pages)
  - Project presentation in the last lecture

    (10%)
  - A project report (maximum1000 words) (10%)
- Suggested information technology topics:
  - Signal processing technologies (e.g., object/audio/image recognition)
  - Advanced digital cellular systems (e.g., 6G)
  - Wi-Fi (e.g., Wi-Fi 6)
  - Satellite communications (e.g., satellite call)
  - Smart systems (e.g., smart city, smart scan, smart robots)
  - …

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Team Project Presentation Marking Scheme

PPT Slides

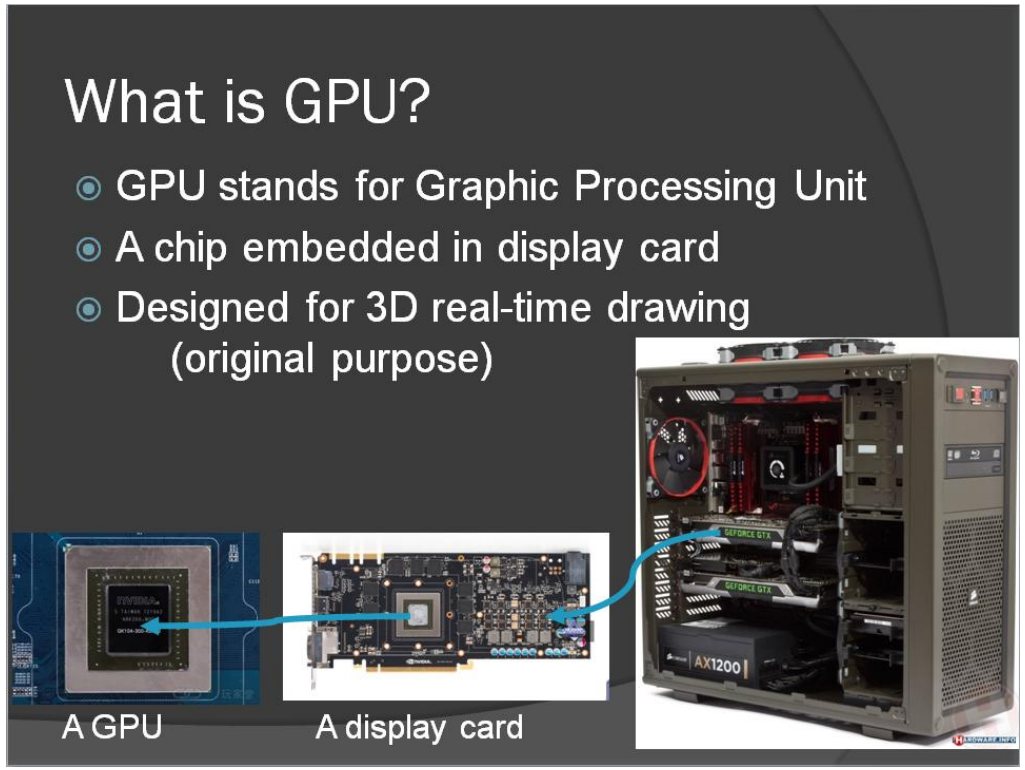
Element \ Score	3	2	1	0
Language and Visual Aid	The language is clear and correct. All the visual aids are clear and legible, and relevant to the content.	The language is clear and correct. Most of the visual aids are clear and legible, and relevant to the content.	The language is mostly clear and correct. Few of the visual aids are clear and legible, and relevant to the content.	The language is not clear and correct. All the visual aids are not clear and irrelevant to the content.
Structure and Organization	All the slides have good layout. The total number of slides is 10-15.	Most of the slides have good layout. Or the total number of slides is not 10-15.	Few of the slides have good layout. And the total number of slides is not 10-15.	The layout of the slides is poor. And the total number of slides is not 10-15.
Content	The content covers all the relevant techniques and background. The contents are presented in a logical order.	The content covers most of the relevant techniques and background. Most of the contents are presented in a logical order.	The content covers few of the relevant techniques and background. Few of the contents are presented in a logical order.	The content does not cover the relevant techniques and background. The contents are not presented in a logical order.
Oral Presentation	The project is clearly presented. The presenter engaged with the audience. The presenter spoke at an appropriate volume, with appropriate body language.	The presenter achieved any two of the following elements: The project is clearly presented. The presenter engaged with the audience. The presenter spoke at an appropriate volume, with appropriate body language	The presenter achieved any one of the following elements: The project is clearly presented. The presenter engaged with the audience. The presenter spoke at an appropriate volume, with appropriate body language	The project is not clearly presented. The presenter did not engage with the audience. The presenter did not speak at an appropriate volume, without appropriate body language.

Team Project Report Marking Scheme

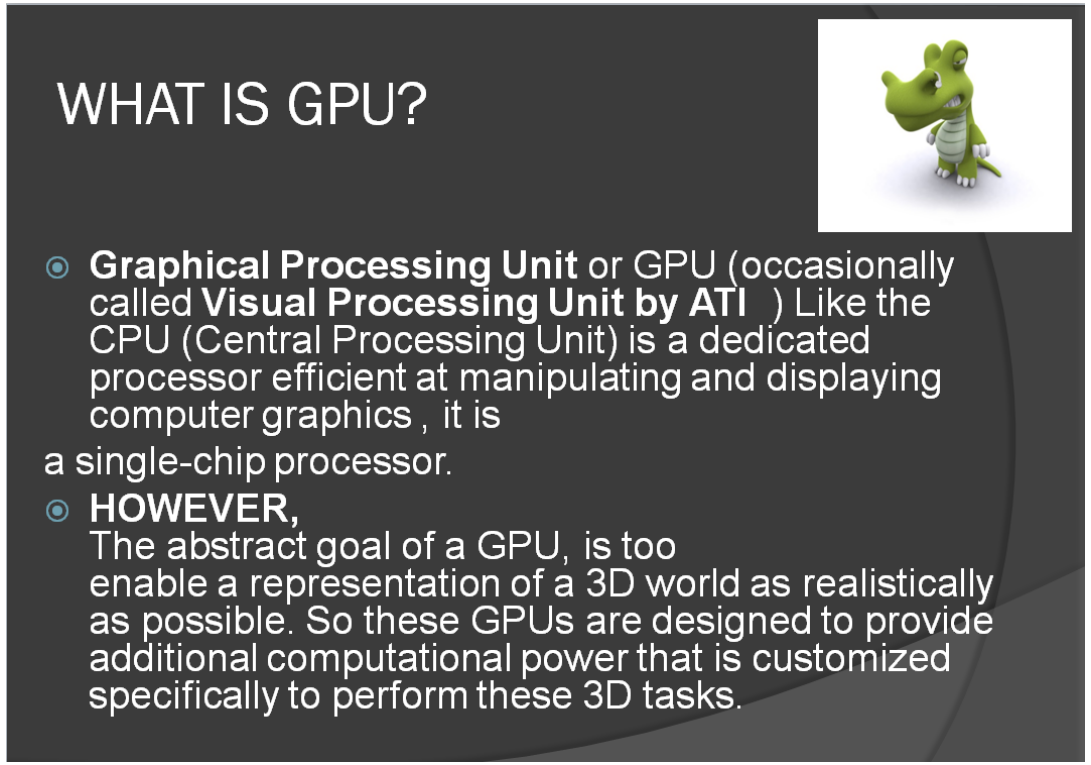
Element \ Score	3	2	1	0
Language and Visual Aid	The language is clear and correct. All the visual aids are clear and legible, and relevant to the content.	The language is clear and correct. Most of the visual aids are clear and legible, and relevant to the content.	The language is mostly clear and correct. Few of the visual aids are clear and legible, and relevant to the content.	The language is not clear and correct. All the visual aids are not clear and irrelevant to the content.
Structure and organization	The report is well organized with the introduction, the main body, and the summary sections.	The introduction or the summary section is missing.	Both of the introduction and the summary sections are missing. Or, the main body section is missing.	The report only contains the introduction or the conclusion section.
Content	The content covers all the relevant techniques and background. The content are presented in a logical order.	The content covers most of the relevant techniques and background. Most of the content are presented in a logical order.	The content covers few of the relevant techniques and background. Few of the content are presented in a logical order.	The content does not cover the relevant techniques and background. The content are not presented in a logical order.
Citation Note: for AI-generated citation, the writer should follow the format below and append the screen shot of the question and AI-generated response to the question. 1. In-text citation: ([Citation number]) 2. Reference list: [Citation number] ChatGPT, response to author query. OpenAI [Online]. https://chatgpt.pro/ (accessed February 15, 2023)	The writing meticulously adheres to the Institute for Electrical and Electronics Engineers (IEEE) citation style, consistently and accurately citing all sources used in the text and references section. In-text citations follow the proper IEEE format, including author names, publication years, and page numbers when necessary. The references section is comprehensive and properly formatted, with all required elements, such as author names, publication dates, titles, and source information, presented in accordance with IEEE guidelines.	The writing consistently adheres to the Institute for Electrical and Electronics Engineers (IEEE) citation style, accurately citing sources used in the text and references section. In-text citations generally follow the proper IEEE format, including author names and publication years. The references section is comprehensive and properly formatted, with all required elements, such as author names, publication dates, titles, and source information, mostly presented in accordance with IEEE guidelines.	The writing partially adheres to the Institute for Electrical and Electronics Engineers (IEEE) citation style, with occasional errors or inconsistencies in the citation format. In-text citations may not consistently follow the proper IEEE format for author names, publication years, or page numbers. The references section may have some omissions or inaccuracies in the required elements, such as author names, publication dates, titles, or source information.	The writing does not adhere to the Institute for Electrical and Electronics Engineers (IEEE) citation style, with numerous errors or inconsistencies in the citation format. In-text citations do not follow the proper IEEE format for author names, publication years, or page numbers. The references section may be incomplete, inaccurately formatted, or missing required elements, such as author names, publication dates, titles, or source information.

Visual Aids

- Visual aids are visual materials, such as pictures, charts, and diagrams, that help people understand and remember information.

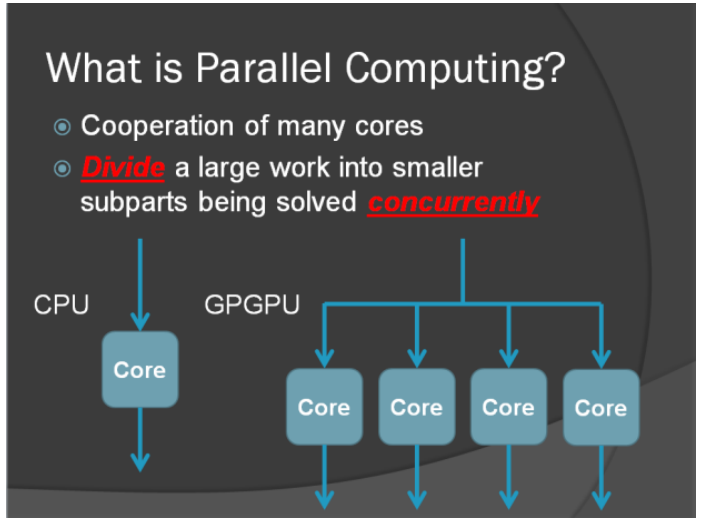


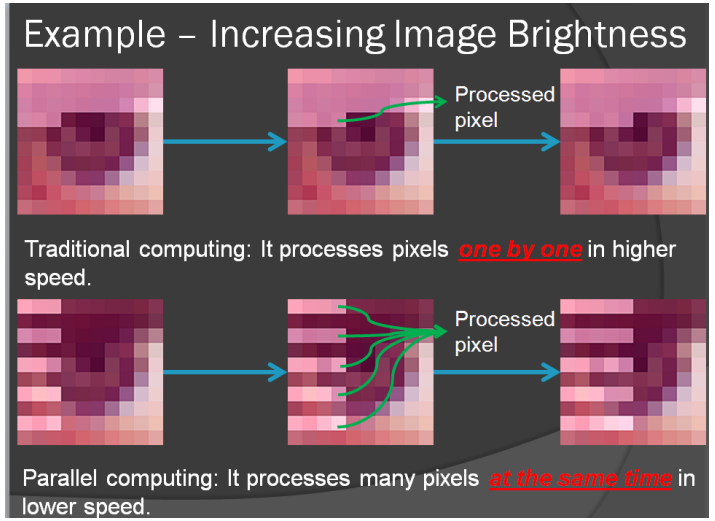
Well used

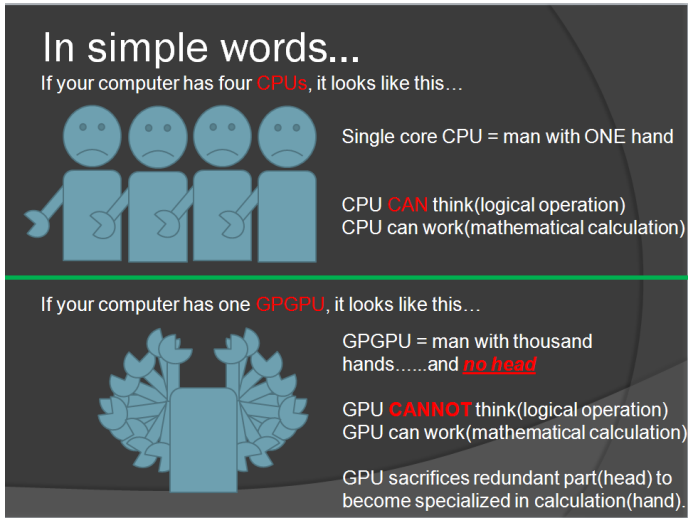


Inappropriate use
Is GPU the animal in the image?

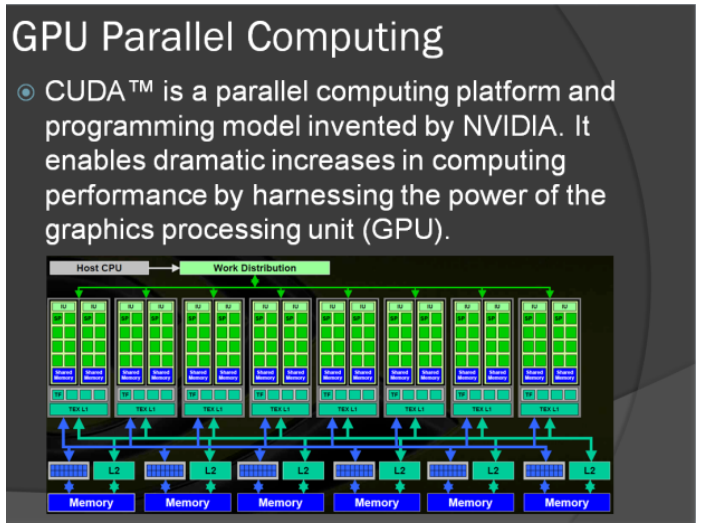
Visual Aids

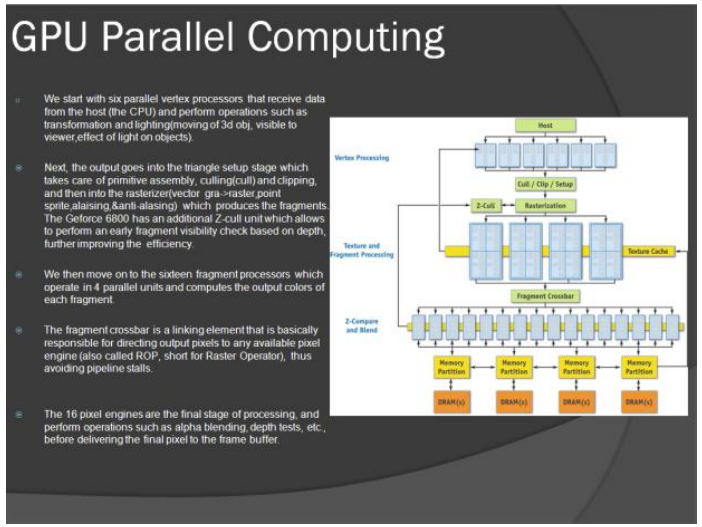


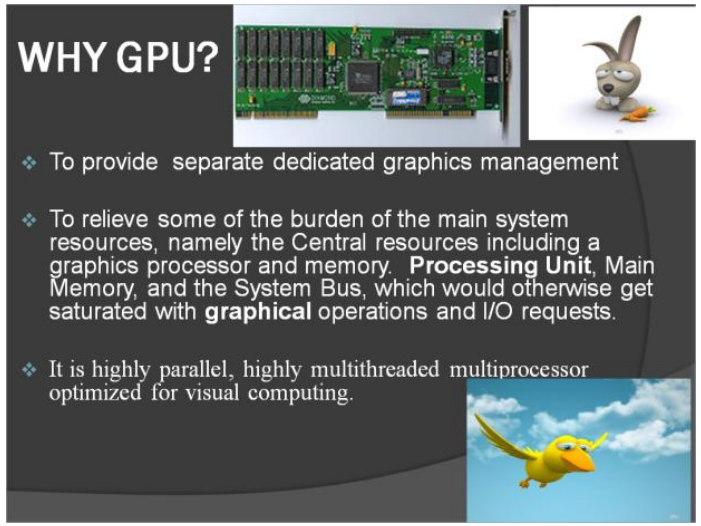




Well used

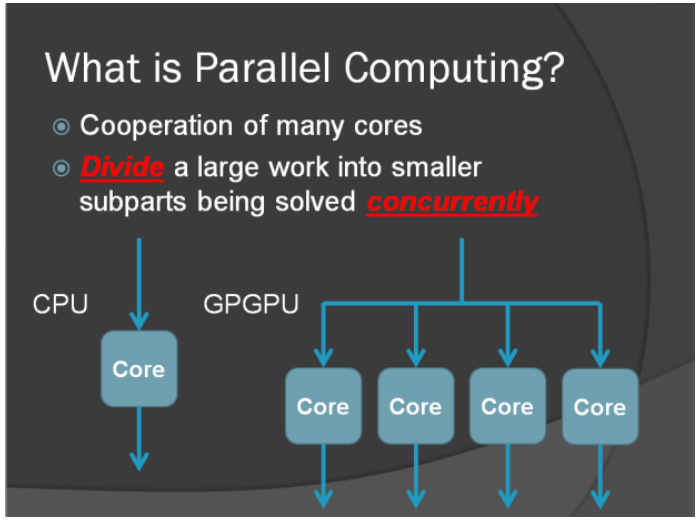


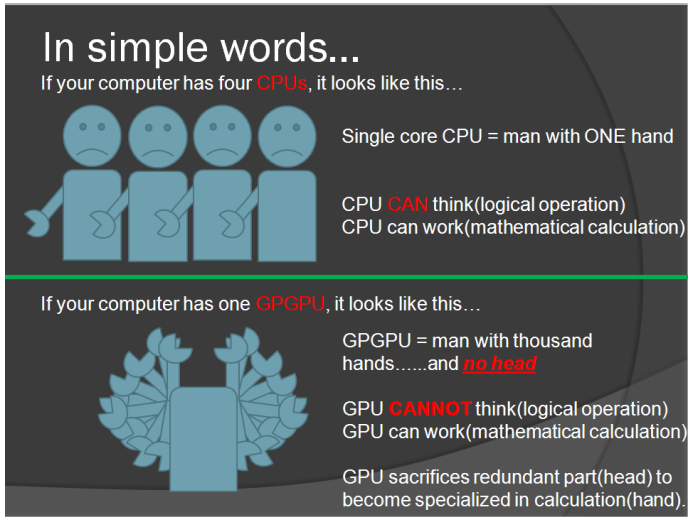




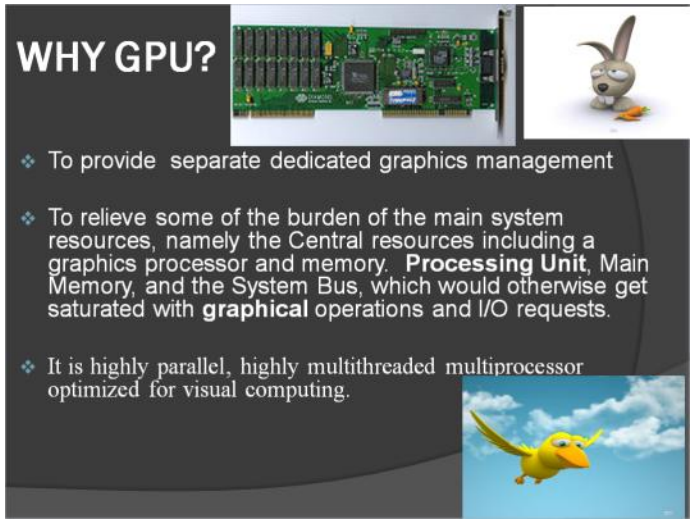
It is not easy to see the direct interaction of the figures and the text in the slides.

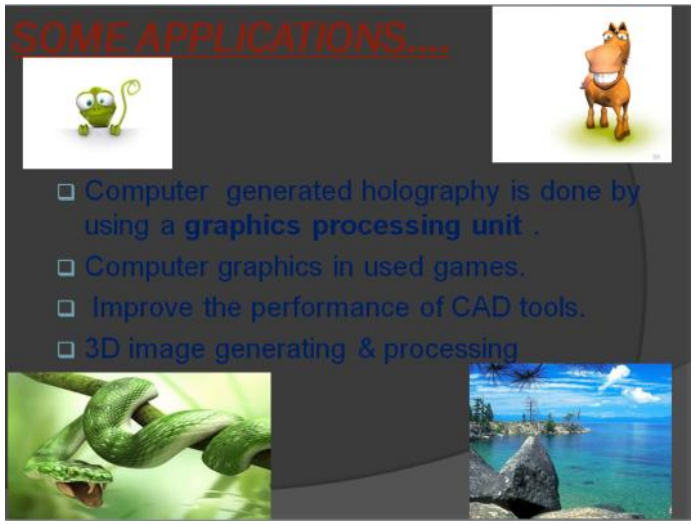
Structure – PowerPoint Presentation Slides





Good layout
Contents in these two slides are arranged in a relatively clear way.





Poor layout
The texts and pictures are not arranged properly.

Structure – Report

1. Introduction

- The first paragraph
- Shorter in length
- Briefly introduce the topic and its background

2. Main body

- The nucleus of the report
- Presents the details of techniques
- Composed of many different paragraphs
- Every paragraph relates to the topic
- Each paragraph focuses on a different piece of the topic

3. Summary

- The last paragraph
- Shorter in length
- Summarizes the main points

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Intended Learning Outcomes

- On successful completion of this course, you will be able to
  - Describe the key technological developments of signals and information technology
  - Identify the fundamental principles related to signals and information technology
  - Solve simple engineering problems related to signals and information technology, such as audio signal processing and modulation, using software tools
  - Create an interactive presentation on state-of-the-art signals and information technology, and to document analysis, evaluation, and/or imagination of a state-of-the-art signals and information technology in a written report
  - Collaborate and communicate effectively in a team setting

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Supplemental Readings

- Huawei Connect 2022: https://weibo.com/7476889887/M7Sb2ayOB
- The impact of information technology (IT): https://master-iesc-angers.com/impacts-of-information-technology-it/

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